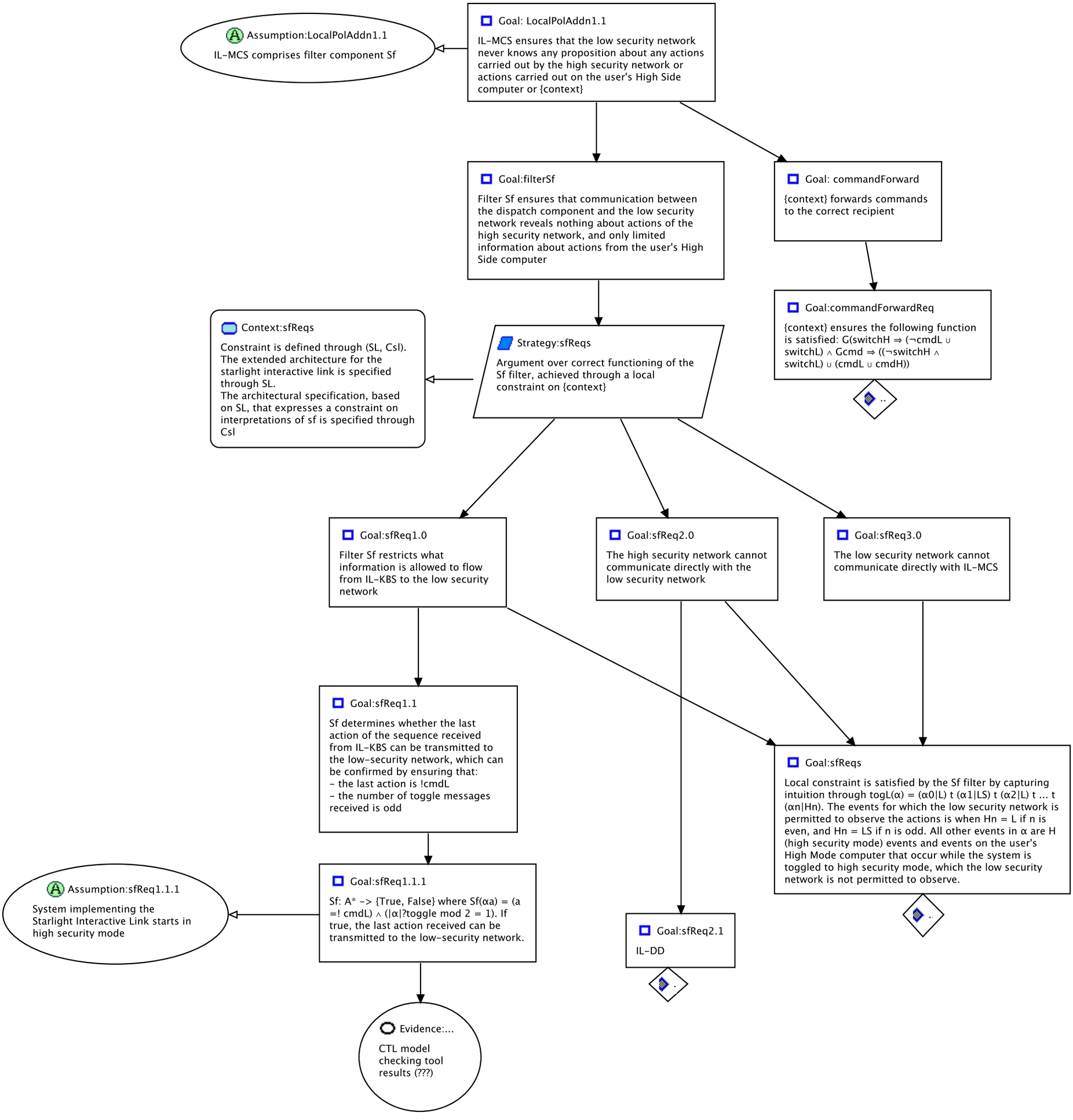
Assumption:LocalPolAddn1.1
IL-MCS comprises filter component Sf
Goal: LocalPolAddn1.1
IL-MCS ensures that the low security network never knows any proposition about any actions carried out by the high security network or actions carried out on the user's High Side computer or {context}
Goal:filterSf
Filter Sf ensures that communication between the dispatch component and the low security network reveals nothing about actions of the high security network, and only limited information about actions from the user's High Side computer
Goal: commandForward
{context} forwards commands to the correct recipient
Goal:commandForwardReq
{context} ensures the following function is satisfied: G(switchH ⇒ (¬cmdL ∪ switchL) ∧ Gcmd ⇒ ((¬switchH ∧ switchL) ∪ (cmdL ∪ cmdH))
Context:sfReqs
Constraint is defined through (SL, Csl). The extended architecture for the starlight interactive link is specified through SL.
The architectural specification, based on SL, that expresses a constraint on interpretations of sf is specified through Csl
Strategy:sfReqs
Argument over correct functioning of the Sf filter, achieved through a local constraint on {context}
Goal:sfReq1.0
Filter Sf restricts what information is allowed to flow from IL-KBS to the low security network
Goal:sfReq2.0
The high security network cannot communicate directly with the low security network
Goal:sfReq3.0
The low security network cannot communicate directly with IL-MCS
Goal:sfReq1.1
Sf determines whether the last action of the sequence received from IL-KBS can be transmitted to the low-security network, which can be confirmed by ensuring that:
- the last action is !cmdL
- the number of toggle messages received is odd
Goal:sfReqs
Local constraint is satisfied by the Sf filter by capturing intuition through togL(α) = (α0|L) t (α1|LS) t (α2|L) t ... t (αn|Hn). The events for which the low security network is permitted to observe the actions is when Hn = L if n is even, and Hn = LS if n is odd. All other events in α are H (high security mode) events and events on the user's High Mode computer that occur while the system is toggled to high security mode, which the low security network is not permitted to observe.
Assumption:sfReq1.1.1
System implementing the Starlight Interactive Link starts in high security mode
Goal:sfReq1.1.1
Sf: A* -> {True, False} where Sf(αa) = (a =! cmdL) ∧ (|α|?toggle mod 2 = 1). If true, the last action received can be transmitted to the low-security network.
Goal:sfReq2.1
IL-DD
Evidence:...
CTL model checking tool results (???)